

step2_preprocessing

November 8, 2025

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[7]: #Import libraries and load temp data

import pandas as pd
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.impute import SimpleImputer
import joblib
```

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[8]: train = pd.read_csv("Load_model/train_temp.csv")
live = pd.read_csv("Load_model/live_temp.csv")

print("Train shape:", train.shape)
print("Live shape:", live.shape)
```

Train shape: (20900, 19)

Live shape: (6967, 18)

```
[9]: #Define features and target

X = train.drop(columns=['label']) # Features
y = train['label']                # Target

print("Features shape:", X.shape)
print("Target shape:", y.shape)
```

Features shape: (20900, 18)

Target shape: (20900,)

```
[10]: #Identify numerical and categorical columns

num_cols = X.select_dtypes(include=['int64', 'float64']).columns.tolist()
cat_cols = X.select_dtypes(include=['object', 'category']).columns.tolist()

print("Numerical columns:", num_cols)
print("Categorical columns:", cat_cols)
```

Numerical columns: ['id', 'social-security-number', 'house-number', 'age-group',

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'fnlwgt', 'education-num', 'capitalgain', 'capitalloss', 'hoursperweek']
Categorical columns: ['workclass', 'education', 'marital-status', 'occupation',
'relationship', 'race', 'sex', 'native-country-code', 'native-country']
```

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[11]: #Build preprocessing pipelines

numeric_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='mean')),
    ('scaler', StandardScaler())
])

categorical_transformer = Pipeline(steps=[
    ('imputer', SimpleImputer(strategy='most_frequent')),
    ('encoder', OneHotEncoder(handle_unknown='ignore'))
])

preprocessor = ColumnTransformer(transformers=[
    ('num', numeric_transformer, num_cols),
    ('cat', categorical_transformer, cat_cols)
])

print("Preprocessor created.")
```

Preprocessor created.

```
[12]: #Save X, y, and preprocessor for next step

X.to_csv("preprocess_data/X_temp.csv", index=False)
y.to_csv("preprocess_data/y_temp.csv", index=False)
joblib.dump(preprocessor, "preprocess_data/preprocessor.pkl")

print("Saved X_temp.csv, y_temp.csv, and preprocessor.pkl.")
```

Saved X_temp.csv, y_temp.csv, and preprocessor.pkl.